DISEASE NOTE OPEN

First Report of the Barley Root-Knot Nematode, *Meloidogyne naasi*, from Pennsylvania, with New Host Report

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Meloidogyne naasi Franklin, 1965, the barley root-knot nematode, was originally found in field crops such as cereals, grasses, and sugar beet (*Beta vulgaris* L.) in England and Wales (Franklin 1965). This nematode is one of the most significant root-knot nematodes impacting grains in European countries (Santos et al. 2020). Among root-knot nematode species, *M. naasi* exhibits a distinct preference for grasses, with documented impacts on turfgrasses, leading to reduced growth and vigor (Skantar et al. 2023; Cook and Yeates 1993). In September 2022, root-knot nematode females and second-stage juveniles (J2) were recovered from roots of fowl manna grass, *Glyceria striata* (Lam.) Hitchc., during a nematode survey on natural vegetation at the Allegheny National Forest (41°30′13.8″N, 79°09′46.2″W). Second-stage juvenile specimens were recovered from soil using sugar centrifugal flotation (Jenkins 1964). Small galls with egg masses were dissected from fowl manna grass roots originally collected from the surveyed areas. In parallel, five plants of noninfected fowl manna grass were placed in a pot in the greenhouse using naturally nematode-infested soil collected from the same forested area. Small galls and female specimens recovered from these plants were dissected and processed for further analyses. Female and J2 were fixed in 3% formaldehyde solution and processed to glycerin (Golden 1990; Hooper 1970). The specimens were examined by light microscopy, morphometric measurements, and molecular markers, which included the D2-D3 region of the large ribosomal subunit 28S and the rDNA internal transcribed spacer region (ITS). The perineal patterns of five females analyzed morphologically were consistent to the patterns observed for *M. naasi*. The perineal patterns had coarse ridges on the cuticle in the dorsal region, forming broken irregular lines around the anal and phasmid area. We also noted a prominent fold that covered some of the anus and showed a curved line between the vulval slit and phasmids, typical of *M. naasi*. The area around the vulval area had a few or no striae except for a few lines radiating from the vulval slit as in the original description. Measurements of 10 J2 had a body length ranged between 380 and 410 µm, stylet 11 to 13 µm, tail 50 to 70 µm long with a hyaline tail terminus between 12 and 22 µm in length, 4 lines in the lateral field, a and c ratio between 29.23 to 35.91 and 5.79 to 7.9, fitting the original description by Franklin, 1965, and other populations found in the United States (Skantar et al. 2023). The matrix codes for the female specimens are A32, B324, C3, and D3, and for J2s, A2, B21, C123, D1, E3, and F12 (Subbotin et al. 2021). The amplified DNA fragments were sequenced, resulting in a 726-bp fragment flanked by the D2-D3 primers (PP097762), while for the ITS primers, a 634-bp fragment was obtained (PP092043). Both generated sequences for the specimens collected in Pennsylvania revealed >99% similarity to *M. naasi* sequences deposited at GenBank, thus validating the morphological analyses. Based on both morphological and molecular analyses, the specimens collected in the state of Pennsylvania were identified as *M. naasi.* To our knowledge, this is the first report of this species from this state and being associated with naturally infected fowl manna grass.

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